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Battery, power consumption device, method and device for preparing a battery

Abstract

Embodiments of the present application provide a battery. The battery includes: a battery cell comprising a pressure relief mechanism, the pressure relief mechanism being arranged in a first wall of the battery cell and the pressure relief mechanism being configured, when an internal pressure or temperature of the battery cell reaches a threshold, to be actuated to release the internal pressure; and a thermal management component for containing a fluid to adjust the temperature of the battery cell; wherein a first surface of the thermal management component is attached to the first wall, and the thermal management component is configured to be capable of being damaged when the pressure relief mechanism is actuated, so that the fluid is discharged from inside of the thermal management component. According to the technical solutions of the embodiments of the present application, the safety of the battery can be enhanced.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/113,038 filed on Dec. 5, 2020, which is a continuation of International Application No. PCT/CN2020/101437, filed on Jul. 10, 2020, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Embodiments of the application relate to the field of batteries, and more particularly to a battery, a power consumption device, and a method and a device for preparing a battery.

BACKGROUND

(2) Energy saving and emission reduction are the key to the sustainable development of the automobile industry. In this case, electric vehicles have become an important part of the sustainable development of the automobile industry because of their advantages of energy saving and environmental friendliness. For the electric vehicles, the battery technology is an important factor related to their development.

(3) In the development of the battery technology, in addition to improving the performance of batteries, safety is also an issue that cannot be ignored. If the safety of the batteries cannot be ensured, the batteries cannot be used. Therefore, how to enhance the safety of the batteries is an urgent technical problem to be solved in the battery technology.

SUMMARY

(4) Embodiments of the application provides a battery, a power consumption device, a method and a device for preparing a battery, which can enhance the safety of the battery.

(5) In a first aspect, a battery is provided, comprising: a battery cell comprising a pressure relief mechanism, the pressure relief mechanism being arranged in a first wall of the battery cell and the pressure relief mechanism being configured, when an internal pressure or temperature of the battery cell reaches a threshold, to be actuated to release the internal pressure; and a thermal management component for containing a fluid to adjust a temperature of the battery cell; wherein a first surface of the thermal management component is attached to the first wall, and the thermal management component is configured to be capable of being damaged when the pressure relief

mechanism is actuated, so that the fluid is discharged from inside of the thermal management component.

(6) In an embodiment of the present application, the first surface of the thermal management component is attached to the first wall provided with the pressure relief mechanism therein. As such, when the pressure relief mechanism is actuated, emissions from the battery cell are discharged toward the thermal management component. Also, the thermal management component is configured to be capable of being damaged when the pressure relief mechanism is actuated, so that a fluid is discharged from the inside of the thermal management component. As such, the fluid may absorb the heat from the battery cell, and reduce the temperature of the emissions. Due to the cooling of the fluid, the temperature of the emissions from the battery cell may be reduced rapidly, so that the risk caused by the abnormality of a single battery cell can be suppressed in the first time, the possibility of battery explosion can be reduced, and the safety of the battery can be enhanced.

(7) In some embodiments, the thermal management component is configured to be capable of being damaged by emissions discharged from the battery cell when the pressure relief mechanism is actuated, so that the emissions pass through the thermal management component.

(8) With the actuation of the pressure relief mechanism, the emissions discharged from the battery cell pass through the thermal management component and quickly move away from the battery cell, which further reduces the risk caused by the emissions.

(9) In some embodiments, the thermal management component is provided with a recess therein, a side surface of the recess is configured to be capable of being damaged by emissions discharged from the battery cell, so that the fluid is discharged from the inside of the thermal management component.

(10) In some embodiments, a bottom wall of the recess is configured to be capable of being damaged by the emissions when the pressure relief mechanism is actuated, so that the emissions pass through the thermal management component.

(11) In some embodiments, the side surface of the recess is configured to be capable of being broken through and/or melted by the emissions, so that the fluid is discharged from the inside of the thermal management component.

(12) In some embodiments, the bottom wall of the recess is configured to be capable of being broken through and/or melted by the emissions, so that the emissions pass through the thermal management component.

(13) In the case of use of the recess, when the pressure relief mechanism is actuated, the emissions from the battery cell rush into the recess. Since the bottom wall of the recess is relatively weak, the emissions will damage the bottom wall of the recess and enter a collection chamber. In addition, the emissions rushing into the recess also melt the side face of the recess, so that the fluid is discharged from the inside of the thermal management component, thereby cooling the hot emissions.

(14) In some embodiments, a radial dimension of the recess gradually decreases in a direction away from the pressure relief mechanism. This can increase the contact area with the emissions and facilitate the damage by the emissions.

(15) In some embodiments, the bottom wall of the recess is provided with a weakened zone, the weakened zone being configured to be capable of being damaged by the emissions when the pressure relief mechanism is actuated, so that the emissions pass through the weakened zone.

(16) The provision of the weakened zone facilitates the emissions passing through the thermal management component.

(17) In some embodiments, the weakened zone has a thickness less than or equal to 3 mm.

(18) In some embodiments, the weakened zone has a lower melting point than the rest of the thermal management component.

(19) In some embodiments, the material of the weakened zone has a melting point below 400° C.

(20) In some embodiments, the recess is provided in the first surface.

(21) In some embodiments, the thermal management component comprises a first thermally

conductive plate and a second thermally conductive plate, the first thermally conductive plate is located between the first wall and the second thermally conductive plate and attached to the first wall, a first region of the first thermally conductive plate is recessed toward the second thermally conductive plate to form the recess, and the first region is connected to the second thermally conductive plate.

(22) In some embodiments, the first region is provided with a first through hole, and a radial dimension of the first through hole is smaller than that of the recess.

(23) In some embodiments, a thickness of the second thermally conductive plate corresponding to the first through hole is smaller than that of the second thermally conductive plate in other regions. In this way, the weakened zone is more easily damaged by the emissions.

(24) In some embodiments, the recess is configured as an avoidance chamber for enabling the pressure relief mechanism to be opened when the pressure relief mechanism is actuated.

(25) The avoidance chamber provides a deformation space for the pressure relief mechanism, such that the pressure relief mechanism is deformed toward the thermal management component and fractured.

(26) In some embodiments, a depth of the recess is related to a size of the pressure relief mechanism.

(27) In some embodiments, the recess has a depth greater than 1 mm.

(28) In some embodiments, the area of an opening of the recess is related to the area of the pressure relief mechanism.

(29) In some embodiments, a ratio of an area of the opening of the recess to an area of the pressure relief mechanism ranges from 0.5 to 2.

(30) In some embodiments, at least a portion of the pressure relief mechanism protrudes outward from the first wall, and the avoidance chamber is configured to accommodate the at least portion of the pressure relief mechanism.

(31) In some embodiments, a portion of the first wall around the pressure relief mechanism protrudes outward, and the avoidance chamber is configured to accommodate an outward protruding portion of the first wall around the pressure relief mechanism.

(32) In this way, the first wall of the battery cell can be closely attached to the surface of the thermal management component, which facilitates the fixation of the battery cell and can also save space and improve the thermal management efficiency. Moreover, when the pressure relief mechanism is actuated, the emissions from the battery cell can be discharged toward the avoidance chamber and away from the battery cell, thereby reducing the risk resulting from the emissions, so that the safety of the battery can be enhanced.

(33) In some embodiments, the thermal management component is provided with a second through hole therein, the second through hole being configured such that emissions discharged from the battery cell can pass through the thermal management component via the second through hole when the pressure relief mechanism is actuated.

(34) In some embodiments, a hole wall of the second through hole is configured to be capable of being damaged by the emissions, so that the fluid is discharged from the inside of the thermal management component.

(35) In some embodiments, the hole wall of the second through hole is configured to be capable of being broken through and/or melted by the emissions, so that the fluid is discharged from the inside of the thermal management component.

(36) When the pressure relief mechanism is actuated, the emissions from the battery cell rush into the second through hole. Since the emissions have high pressure and high temperature, the emissions further melt the hole wall of the second through hole when passing through the second through hole, so that the fluid is discharged from the inside of the thermal management component, thereby cooling the emissions.

(37) In some embodiments, a radial dimension of the second through hole gradually decreases in a

direction away from the pressure relief mechanism. This can increase the contact area with the emissions and facilitate the damage by the emissions.

(38) In some embodiments, an area of an opening of the second through hole is related to an area of the pressure relief mechanism.

(39) In some embodiments, a ratio of the area of the opening of the second through hole to the area of the pressure relief mechanism ranges from 0.5 to 2.

(40) In some embodiments, at least a portion of the pressure relief mechanism protrudes from the first wall, and the second through hole is used to accommodate the at least portion of the pressure relief mechanism.

(41) In this way, the first wall of the battery cell can be closely attached to the surface of the thermal management component, which facilitates the fixation of the battery cell and can also save space and improve the thermal management efficiency. Moreover, when the pressure relief mechanism is actuated, the emissions from the battery cell can be discharged toward the second through hole and away from the battery cell, thereby reducing the risk resulting from the emissions, so that the safety of the battery can be enhanced.

(42) In some embodiments, a portion of the first wall around the pressure relief mechanism protrudes outward, and the second through hole is used to accommodate an outward protruding portion of the first wall around the pressure relief mechanism.

(43) In some embodiments, the pressure relief mechanism is provided with a breaking device, the breaking device being used to damage the thermal management component when the pressure relief mechanism is actuated, so that the fluid is discharged from the inside of the thermal management component.

(44) In some embodiments, the breaking device is a spike.

(45) In some embodiments, the battery further comprises: an electrical chamber for accommodating a plurality of the battery cells; and a collection chamber for collecting emissions discharged from the battery cells and emissions from the thermal management component when the pressure relief mechanism is actuated; wherein the thermal management component is configured to isolate the electrical chamber from the collection chamber.

(46) The electrical chamber for accommodating the battery cells is separated from the collection chamber for collecting the emissions by means of the thermal management component. When the pressure relief mechanism is actuated, the emissions from the battery cell enter the collection chamber, rather than the electrical chamber or with a little thereof entering the electrical chamber, such that the electrical connections in the electrical chamber will not be affected, and the safety of the battery thus can be enhanced.

(47) In some embodiments, the thermal management component has a wall shared by the electrical chamber and the collection chamber.

(48) Since the thermal management component has the wall shared by the electrical chamber and the collection chamber, the emissions can be isolated from the electrical chamber as far as possible, thus reducing the risk resulting from the emissions and enhancing the safety of the battery.

(49) In some embodiments, the battery further comprises: a protective member, the protective member being configured to protect the thermal management component, and the protective member and the thermal management component forming the collection chamber.

(50) The collection chamber formed by the protective member and the thermal management component can effectively collect and buffer the emissions and reduce the risk resulting therefrom.

(51) In some embodiments, the electrical chamber is isolated from the collection chamber by the thermal management component.

(52) The collection chamber is not in communication with the electrical chamber, and liquid or gas, etc. in the collection chamber cannot enter the electrical chamber, so that the electrical chamber can be better protected.

(53) In some embodiments, the thermal management component is configured to enable the

emissions discharged from the battery cell to pass through the thermal management component and enter the collection chamber when the pressure relief mechanism is actuated.

(54) In some embodiments, a second wall of the battery cell is provided with an electrode terminal, and the second wall is different from the first wall.

(55) The pressure relief mechanism and the electrode terminals are provided on different walls of the battery cell, such that when the pressure relief mechanism is actuated, the emissions from the battery cell can be farther away from the electrode terminals, thereby reducing the impact of the emissions on the electrode terminals and a bus component and therefore enhancing the safety of the battery.

(56) In some embodiments, the second wall is arranged opposite to the first wall.

(57) In some embodiments, the pressure relief mechanism is a temperature-sensitive pressure relief mechanism configured to be capable of being melted when an internal temperature of the battery cell reaches a threshold; and/or the pressure relief mechanism is a pressure-sensitive pressure relief mechanism configured to be capable of being ruptured when an internal pressure of the battery cell reaches a threshold.

(58) In a second aspect, a power consumption device is provided, comprising: a battery of the first aspect.

(59) In some embodiments, the power consumption device is a vehicle, a ship or a spacecraft.

(60) In a third aspect, a method for preparing a battery is provided, comprising: providing a battery cell comprising a pressure relief mechanism, the pressure relief mechanism being arranged in a first wall of the battery cell and the pressure relief mechanism being configured, when an internal pressure or temperature of the battery cell reaches a threshold, to be actuated to release the internal pressure; providing a thermal management component, the thermal management component being used to contain a fluid; and attaching a first surface of the thermal management component to the first wall, wherein the thermal management component can be damaged when the pressure relief mechanism is actuated, so that the fluid is discharged from inside of the thermal management component.

(61) In some embodiments, the thermal management component can be damaged by emissions discharged from the battery cell when the pressure relief mechanism is actuated, so that the emissions pass through the thermal management component.

(62) In some embodiments, the thermal management component is provided with a recess therein, and a bottom wall of the recess can be damaged by the emissions discharged from the battery cell when the pressure relief mechanism is actuated, so that the emissions pass through the thermal management component, and the side surface of the recess can be damaged by the emissions, so that the fluid is discharged from the inside of the thermal management component.

(63) In some embodiments, the bottom wall of the recess is provided with a weakened zone, the weakened zone being configured to be capable of being damaged by emissions when the pressure relief mechanism is actuated, so that the emissions pass through the weakened zone.

(64) In some embodiments, the thermal management component is provided with a second through hole, the emissions discharged from the battery cell can pass through the thermal management component via the second through hole when the pressure relief mechanism is actuated, and a hole wall of the second through hole can be damaged by the emissions, so that the fluid is discharged from inside of the thermal management component.

(65) In a fourth aspect, a device for preparing a battery is provided, comprising a module for carrying out the method of the third aspect described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings described herein are intended to provide a further understanding of the present application, which constitute a part of the present application. The illustrative embodiments of the present application and the description thereof are for explaining the present application and do not constitute an undue limitation of the present application. In the drawings:
- (2) FIG. 1 is a schematic diagram of a vehicle according to an embodiment of the present application;
- (3) FIG. 2 is a schematic structural diagram of a battery according to an embodiment of the present application;
- (4) FIG. 3 is a schematic structural diagram of a battery module according to an embodiment of the present application;
- (5) FIG. 4 is an exploded view of a battery cell according to an embodiment of the present application;
- (6) FIG. 5 is an exploded view of a battery cell according to another embodiment of the present application;
- (7) FIGS. 6 and 7 are schematic structural diagrams of the structure of the battery according to some embodiments of the present application;
- (8) FIG. 8a is a schematic plan view of the battery according to an embodiment of the present application;
- (9) FIG. 8b is a schematic diagram of a cross section of the battery shown in FIG. 8a taken along the line A-A;
- (10) FIG. 8c is an enlarged view of a part B of the battery shown in FIG. 8b;
- (11) FIG. 9a is a schematic perspective view of a thermal management component according to an embodiment of the present application;
- (12) FIG. 9b is a schematic cross-sectional diagram of the thermal management component of FIG. 9a taken along A-A;
- (13) FIG. 9c is an exploded view of a thermal management component according to an embodiment of the present application;
- (14) FIGS. 10-17 are schematic structural diagrams of the battery according to some embodiments of the present application;
- (15) FIG. 18 is an exploded view of a battery according to an embodiment of the present application;
- (16) FIG. 19 is a schematic flowchart of a method for preparing a battery according to an embodiment of the present application; and
- (17) FIG. 20 is a schematic block diagram of a device for preparing a battery according to an embodiment of the present application.

DESCRIPTION OF EMBODIMENTS

- (18) In order to make objects, technical solutions and advantages of embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be clearly described below with reference to the drawings for the embodiments of the present application. Apparently, the described embodiments are some of, rather than all of, the embodiments of the present application. All the other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present application without any creative effort shall fall within the scope of protection of the present application.
- (19) Unless otherwise defined, all technical and scientific terms used in the present application have the same meanings as those commonly understood by those skilled in the art to which the present application belongs. The terms used in the specification of the present application are merely for the purpose of describing specific embodiments, but are not intended to limit the present application. The terms “comprising” and “having” and any variations thereof in the specification

and the claims of the present application as well as the foregoing description of the accompanying drawings are intended to cover non-exclusive inclusions. The terms “first”, “second” and the like in the specification and the claims of the present application as well as the above drawings are used to distinguish different objects, rather than to describe a specific order or primary-secondary relationship.

(20) The phrase “embodiments” referred to in the present application means that the descriptions of specific features, structures, and characteristics in combination with the embodiments are included in at least one embodiment of the present application. The phrase at various locations in the specification does not necessarily refer to the same embodiment, or an independent or alternative embodiment exclusive of another embodiment. Those skilled in the art understand, in explicit and implicit manners, that an embodiment described in the present application may be combined with another embodiment.

(21) In the description of the present application, it should be noted that unless otherwise explicitly specified and defined, the terms “mounting”, “connecting”, “connection” and “attaching” should be understood in a broad sense, for example, they may be a fixed connection, a detachable connection, or an integrated connection; may be a direct connection and may also be an indirect connection via an intermediate medium, or may be communication between the interiors of two elements. A person of ordinary skill in the art may understand the specific meanings of the foregoing terms in the present application according to specific circumstances.

(22) In the present application, the term “and/or” is only an association relation describing associated objects, which means that there may be three relations, for example, A and/or B may represent three situations: A exists alone, both A and B exist, and B exists alone. In addition, the character “/” in the present application generally indicates that the associated objects before and after the character are in an “or” relation.

(23) In the present application, “a plurality of” means two or more (including two), similarly, “a plurality of groups” means two or more groups (including two groups), and “a plurality of sheets” means two or more sheets (including two sheets).

(24) In the present application, a battery cell may include a lithium-ion secondary battery, a lithium-ion primary battery, a lithium-sulfur battery, a sodium/lithium-ion battery, a sodium-ion battery or a magnesium-ion battery, etc., which is not limited by the embodiments of the present application. The battery cell may be cylindrical, flat, cuboid or in another shape, which is not limited by the embodiments of the present application. The battery cell is generally divided into three types according to the way of packaging: a cylindrical battery cell, a prismatic battery cell and a pouch battery cell, which is not limited by the embodiments of the present application.

(25) A battery mentioned in the embodiments of the present application refers to a single physical module including one or more battery cells to provide a higher voltage and capacity. For example, the battery mentioned in the present application may include a battery module or a battery pack, etc. The battery generally includes a case for enclosing one or more battery cells. The case can prevent liquid or other foreign matters from affecting the charging or discharging of the battery cell.

(26) The battery cell includes an electrode assembly and an electrolytic solution, and the electrode assembly is composed of a positive electrode sheet, a negative electrode sheet and an isolation film. The operation of the battery cell mainly relies on the movement of metal ions between the positive electrode sheet and the negative electrode sheet. The positive electrode sheet includes a positive electrode current collector and a positive electrode active material layer. The positive electrode active material layer is coated on a surface of the positive electrode current collector, and the current collector not coated with the positive electrode active material layer protrudes from the current collector coated with the positive electrode active material layer and is used as a positive electrode tab. Taking a lithium-ion battery as an example, the material of the positive electrode current collector may be aluminum, and the positive electrode active material may be lithium

cobalt oxides, lithium iron phosphate, ternary lithium or lithium manganate, etc. The negative electrode sheet includes a negative electrode current collector and a negative electrode active material layer. The negative electrode active material layer is coated on a surface of the negative electrode current collector, and the current collector not coated with the negative electrode active material layer protrudes from the current collector coated with the negative electrode active material layer and is used as a negative electrode tab. The material of the negative electrode current collector may be copper, and the negative electrode active material may be carbon or silicon, etc. In order to ensure that no fusing occurs when a large current passes, there are a plurality of positive electrode tabs which are stacked together, and there are a plurality of negative electrode tabs which are stacked together. The material of the isolation film may be PP or PE, etc. In addition, the electrode assembly may have a coiled structure or a laminated structure, and the embodiments of the present application are not limited thereto. With the development of the battery technology, it is necessary to consider many design factors, such as energy density, cycle life, discharge capacity, C-rate and other performance parameters. In addition, the safety of the battery should also be considered.

(27) With respect to the battery cell, the main safety hazards come from the charging and discharging processes, and a suitable environmental temperature design is also required. In order to effectively avoid unnecessary losses, at least triple protection measures are generally taken for the battery cell. Specifically, the protection measures include at least a switching element, a properly selected isolation film material and a pressure relief mechanism. The switching element refers to an element that can stop the charging or discharging of the battery when the temperature or resistance in the battery cell reaches a certain threshold. The isolation film is configured to isolate the positive electrode sheet from the negative electrode sheet and can automatically dissolve micron-sized (or even nanoscale) micropores attached to the isolation film when the temperature rises to a certain value, thus preventing metal ions from passing through the isolation film and terminating the internal reaction of the battery cell.

(28) The pressure relief mechanism refers to an element or component that is actuated to release an internal pressure or temperature when the internal pressure or temperature of the battery cell reaches a predetermined threshold. The threshold design is different according to different design requirements. The threshold may depend on the material of one or more of the positive electrode sheet, the negative electrode sheet, the electrolytic solution and the isolation film in the battery cell. The pressure relief mechanism may take the form of an anti-explosion valve, an air valve, a pressure relief valve or a safety valve, etc., and may specifically adopt a pressure-sensitive or temperature-sensitive element or structure. That is, when the internal pressure or temperature of the battery cell reaches a predetermined threshold, the pressure relief mechanism performs an action or a weakened structure provided in the pressure relief mechanism is damaged, so as to form an opening or channel for releasing the internal pressure or temperature.

(29) The “actuation” mentioned in the present application means that the pressure relief mechanism acts or is activated to a certain state, such that the internal pressure and temperature of the battery cell can be released. The action executed by the pressure relief mechanism may include but is not limited to: at least a portion of the pressure relief mechanism being fractured, broken, torn or opened, etc. When the pressure relief mechanism is actuated, high-temperature and high-pressure substances inside the battery cell are discharged outwards from an actuated position as emissions. In this way, the pressure and temperature in the battery cell can be released at a controllable pressure or temperature, thereby avoiding potentially more serious accidents.

(30) The emissions from the battery cell mentioned in the present application include but are not limited to: the electrolytic solution, dissolved or split positive and negative electrode sheets, fragments of the isolation film, high-temperature and high-pressure gas generated by reaction, flame, etc.

(31) The pressure relief mechanism on the battery cell has an important impact on the safety of the

battery. For example, when short circuit, overcharge and other phenomena occur, it may lead to thermal runaway inside the battery cell, resulting in a sudden increase in pressure or temperature. In this case, the internal pressure and temperature can be released outward through the actuation of the pressure relief mechanism, to prevent the battery cell from exploding and catching fire.

(32) In the current design solutions of the pressure relief mechanism, the main concern is to release the high pressure and high temperature inside the battery cell, i.e., to discharge the emissions to the outside of the battery cell. However, in order to ensure an output voltage or current of the battery, a plurality of battery cells are often required and electrically connected to each other via a bus component. The emissions discharged from the interior of a battery cell may cause short circuit of the other battery cells. For example, when discharged metal scraps electrically connect two bus components, the battery may be short-circuited, thereby posing a potential safety hazard. Moreover, the high-temperature and high-pressure emissions are discharged in a direction of the pressure relief mechanism provided in the battery cell, and more specifically, may be discharged in a direction of a region where the pressure relief mechanism is actuated. The strength and destructive power of such emissions may be great, or may even be enough to break through one or more structures in this direction, causing further safety problems.

(33) In view of this, the embodiments of the application provide a technical solution in which a wall of the battery cell provided with the pressure relief mechanism is attached to the thermal management component, and the thermal management component is damaged by means of the actuation of the pressure relief mechanism, so that a fluid inside the thermal management component is discharged and may cool down the emissions to reduce the risk caused by the emissions, thereby enhancing the safety of the battery.

(34) The thermal management component is configured to accommodate a fluid to adjust temperatures of a plurality of battery cells. The fluid here may be liquid or gas, and temperature adjustment means heating or cooling the plurality of battery cells. In the case of cooling or lowering the temperatures of the battery cells, the thermal management component is configured to accommodate a cooling fluid to lower the temperatures of the plurality of battery cells. In this case, the thermal management component may also be called a cooling component, a cooling system or a cooling plate, etc. The fluid accommodated by the thermal management component may also be called a cooling medium or a cooling fluid, and more specifically, may be called a cooling liquid or a cooling gas. In addition, the thermal management component can also be used for heating to raise the temperatures of the plurality of battery cells, which is not limited by the embodiments of the present application. Optionally, the fluid can flow in a circulating manner to achieve better temperature adjustment effects. Optionally, the fluid may be water, a mixture of water and ethylene glycol, or air, etc.

(35) The electrical chamber mentioned in the present application is used to accommodate multiple battery cells and a bus component. The electrical chamber may be sealed or unsealed. The electrical chamber provides an installation space for the battery cells and the bus component. In some embodiments, a structure configured to fix the battery cells may also be provided in the electrical chamber. The shape of the electrical chamber may be determined according to the plurality of battery cells and the bus component which are accommodated therein. In some embodiments, the electrical chamber may be a cube with six walls. Since the battery cells in the electrical chamber form higher voltage output through electrical connection, the electrical chamber may also be called a “high-voltage chamber”.

(36) The bus component mentioned in the present application is used to realize the electrical connection between the plurality of battery cells, such as parallel or series connection or parallel-series connection. The bus component may implement the electrical connection between the battery cells by connecting electrode terminals of the battery cells. In some embodiments, the bus component may be fixed to the electrode terminals of the battery cells by means of welding. Corresponding to the “high-voltage chamber”, the electrical connection formed by the bus

component may also be called “high-voltage connection”.

(37) The collection chamber mentioned in the present application is used to collect the emissions and can be sealed or non-sealed. In some embodiments, the collection chamber may contain air or another gas. In the collection chamber there is no electrical connection to the voltage output. Corresponding to the “high-voltage chamber”, the collection chamber may also be called a “low-voltage chamber”. Optionally, or additionally, the collection chamber may also contain liquid, such as a cooling medium, or a component for accommodating the liquid may also be provided in the collection chamber to further cool the emissions entering the collection chamber. Further, optionally, the gas or the liquid in the collection chamber flows in a circulating manner.

(38) The technical solutions described in the embodiments of the present application are all applicable to various devices using batteries, such as mobile phones, portable apparatuses, notebook computers, electromobility vehicles, electronic toys, electric tools, electric vehicles, ships and spacecrafts. For example, the spacecrafts include airplanes, rockets, space shuttles, spaceships, etc.

(39) It should be understood that the technical solutions described in the embodiments of the present application are not only applicable to the foregoing apparatuses, but also applicable to all apparatuses using batteries. However, for the sake of brevity, the following embodiments take electric vehicles as an example for description.

(40) For example, FIG. 1 is a schematic structural diagram of a vehicle **1** according to an embodiment of the present application. The vehicle **1** may be a fuel vehicle, a gas vehicle or a new-energy vehicle. The new-energy vehicle may be a battery electric vehicle, a hybrid vehicle or an extended-range vehicle, or the like. A motor **40**, a controller **30** and a battery **10** may be provided inside the vehicle **1**, and the controller **30** is configured to control the battery **10** to supply power to the motor **40**. For example, the battery **10** may be provided at the bottom or the head or the tail of the vehicle **1**. The battery **10** may be configured to supply power to the vehicle **1**. For example, the battery **10** can be used as an operation power supply of the vehicle **1** and is used for a circuit system of the vehicle **1**, for example, for a working power demand of the vehicle **1** during startup, navigation and running. In another embodiment of the present application, the battery **10** can be used not only as an operation power supply of the vehicle **1**, but also as a driving power supply of the vehicle **1**, replacing or partially replacing fuel or natural gas to provide driving power for the vehicle **1**.

(41) In order to meet different power requirements, the battery may include a plurality of battery cells, wherein the plurality of battery cells may be in series connection, parallel connection or series-parallel connection. The series-parallel connection refers to a combination of series connection and parallel connection. The battery may also be called a battery pack. Optionally, the plurality of battery cells may be first connected in series, in parallel or in series and parallel to form battery modules, and then the multiple battery modules are connected in series, in parallel or in series and parallel to form a battery. That is, a plurality of battery cells may directly form a battery, or may first form battery modules, and then the battery modules form a battery.

(42) For example, as shown in FIG. 2, which is a schematic structural diagram of a battery **10** according to an embodiment of the present application, the battery **10** may include a plurality of battery cells **20**. The battery **10** may further include a case (or a covering) with the interior thereof being a hollow structure, and the plurality of battery cells **20** are accommodated in the case. As shown in FIG. 2, the case may include two portions, which are referred to as a first portion **111** and a second portion **112**, respectively, and the first portion **111** and the second portion **112** are fastened together. The shapes of the first portion **111** and the second portion **112** may be determined according to the shape of the combined plurality of battery cells **20**, and the first portion **111** and the second portion **112** may each have an opening. For example, the first portion **111** and the second portion **112** each may be a hollow cuboid and each have only one surface with an opening, and the opening of the first portion **111** is arranged opposite to the opening of the second portion **112**. The first portion **111** and the second portion **112** are fastened to each other to form a case with

a closed chamber. The plurality of battery cells **20** are combined in parallel connection or series connection or series-parallel connection and are then placed in the case formed by fastening the first portion **111** to the second portion **112**.

(43) Optionally, the battery **10** may also include other structures, which will not be described in detail herein. For example, the battery **10** may also include a bus component. The bus component is configured to implement the electric connection between the plurality of battery cells **20**, such as parallel connection, series connection or series-parallel connection. Specifically, the bus component may implement the electrical connection between the battery cells **20** by connecting electrode terminals of the battery cells **20**. Further, the bus component may be fixed to the electrode terminals of the battery cells **20** by means of welding. Electric energy of the plurality of battery cells **20** can be further led out through an electrically conductive mechanism passing through the case. Optionally, the electrically conductive mechanism may also belong to the bus component.

(44) According to different power requirements, the number of the battery cells **20** may be set to any value. The plurality of battery cells **20** can be connected in series, in parallel or in series and parallel to implement larger capacity or power. Since there may be many battery cells **20** included in each battery **10**, the battery cells **20** may be arranged in groups for convenience of installation, and each group of battery cells **20** constitutes a battery module. The number of the battery cells **20** included in the battery module is not limited and may be set as required. For example, FIG. 3 shows an example of a battery module. The battery may include a plurality of battery modules, and these battery modules may be connected in series, in parallel or in series and parallel.

(45) FIG. 4 is a schematic structural diagram of a battery cell **20** according to an embodiment of the present application. The battery cell **20** includes one or more electrode assemblies **22**, a housing **211** and a cover plate **212**. The coordinate system shown in FIG. 4 is the same as that in FIG. 3. The housing **211** and the cover plate **212** form a shell or a battery box **21**. A wall of the housing **211** and the cover plate **212** are each referred to as a wall of the battery cell **20**. The housing **211** is shaped according to the shape of one or more electrode assemblies **22** after combination. For example, the housing **211** may be a hollow cuboid or cube or cylinder, and one surface of the housing **211** has an opening such that one or more electrode assemblies **22** can be placed in the housing **211**. For example, when the housing **211** is a hollow cuboid or cube, one plane of the housing **211** is an opening surface, i.e., the plane does not have a wall, so that the inside and outside of the housing **211** are in communication with each other. When the housing **211** is a hollow cylinder, an end face of the housing **211** is an opening surface, i.e., the end face does not have a wall, so that the inside and outside of the housing **211** are in communication with each other. The cover plate **212** covers the opening and is connected to the housing **211** to form a closed cavity in which the electrode assembly **22** is placed. The housing **211** is filled with an electrolyte, such as an electrolytic solution.

(46) The battery cell **20** may further include two electrode terminals **214**, and the two electrode terminals **214** may be provided on the cover plate **212**. The cover plate **212** is generally in the shape of a flat plate, and the two electrode terminals **214** are fixed on a flat plate surface of the cover plate **212**. The two electrode terminals **214** are a positive electrode terminal **214a** and a negative electrode terminal **214b**, respectively. Each electrode terminal **214** is correspondingly provided with a connecting member **23** also called a current collecting member **23**, which is located between the cover plate **212** and the electrode assembly **22** and configured to electrically connect the electrode assembly **22** to the electrode terminal **214**.

(47) As shown in FIG. 4, each electrode assembly **22** has a first electrode tab **221a** and a second electrode tab **222a**. The first electrode tab **221a** and the second electrode tab **222a** have opposite polarities. For example, when the first electrode tab **221a** is a positive electrode tab, the second electrode tab **222a** is a negative electrode tab. The first electrode tab **221a** of one or more electrode assemblies **22** is connected to one electrode terminal via one connecting member **23**, and the second electrode tab **222a** of one or more electrode assemblies **22** is connected to the other

electrode terminal via the other connecting member **23**. For example, the positive electrode terminal **214a** is connected to the positive electrode tab via one connecting member **23**, and the negative electrode terminal **214b** is connected to the negative electrode tab via the other connecting member **23**.

(48) In this battery cell **20**, according to actual use requirements, there may be a single or a plurality of electrode assemblies **22**. As shown in FIG. **4**, there are four separate electrode assemblies **22** in the battery cell **20**.

(49) A schematic structural diagram of a battery cell **20** including a pressure relief mechanism **213** according to another embodiment of the present application is shown in FIG. **5**.

(50) The housing **211**, the cover plate **212**, the electrode assembly **22** and the connecting member **23** in FIG. **5** are consistent with the housing **211**, the cover plate **212**, the electrode assembly **22** and the connecting member **23** in FIG. **4**, which will not be repeated here for brevity.

(51) One wall of the battery cell **20**, such as a first wall **21a** shown in FIG. **5**, may be further provided with a pressure relief mechanism **213**. For convenience of display, the first wall **21a** is separated from the housing **211** in FIG. **5**, but this does not specify that a bottom side of the housing **211** has an opening. The pressure relief mechanism **213** is configured, when an internal pressure or temperature of the battery cell **20** reaches a threshold, to be actuated to release the internal pressure or temperature.

(52) The pressure relief mechanism **213** may be a portion of the first wall **21a** or is split from the first wall **21a** and fixed to the first wall **21a** by means of welding, for example. When the pressure relief mechanism **213** is a portion of the first wall **21a**, e.g., the pressure relief mechanism **213** can be formed by providing an indentation on the first wall **21a**, and the thickness of the first wall **21a** corresponding to the indentation is smaller than that of other regions of the pressure relief mechanism **213** except the indentation. The indentation is the weakest position of the pressure relief mechanism **213**. When excessive gas generated by the battery cell **20** causes the internal pressure of the housing **211** to rise and reach a threshold, or the internal temperature of the battery cell **20** rises and reaches a threshold due to the heat generated by the internal reaction of the battery cell **20**, the pressure relief mechanism **213** can be fractured at the indentation, resulting in the communication between the inside and outside of the housing **211**. The gas pressure and temperature are released outward through the cracking of the pressure relief mechanism **213**, thereby preventing the battery cell **20** from exploding.

(53) Optionally, in an embodiment of the present application, as shown in FIG. **5**, in the case where the pressure relief mechanism **213** is provided at the first wall **21a** of the battery cell **20**, a second wall of the battery cell **20** is provided with electrode terminals **214** and is different from the first wall **21a**.

(54) Optionally, the second wall is arranged opposite to the first wall **21a**. For example, the first wall **21a** may be a bottom wall of the battery cell **20**, and the second wall may be a top wall of the battery cell **20**, i.e., the cover plate **212**.

(55) Optionally, as shown in FIG. **5**, the battery cell **20** may also include a backing plate **24**. The backing plate **24** is located between the electrode assembly **22** and the bottom wall of the housing **211**, can support the electrode assembly **22**, and can also effectively prevent the electrode assembly **22** from interfering with rounded corners around the bottom wall of the housing **211**. In addition, the backing plate **24** may be provided with one or more through holes, e.g., the backing plate may be provided with a plurality of uniformly arranged through holes, or when the pressure relief mechanism **213** is provided on the bottom wall of the housing **211**, through holes are formed at positions corresponding to the pressure relief mechanism **213** for facilitating the guiding of liquid and gas. Specifically, this can communicate spaces of an upper surface and a lower surface of the backing plate **24**, and gas generated inside the battery cell **20** and the electrolytic solution can freely pass through the backing plate **24**.

(56) The pressure relief mechanism **213** and the electrode terminals **214** are provided on different

walls of the battery cell **20**, such that when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** can be farther away from the electrode terminals **214**, thereby reducing the impact of the emissions on the electrode terminals **214** and the bus component and therefore enhancing the safety of the battery.

(57) Further, when the electrode terminals **214** are provided on the cover plate **212** of the battery cell **20**, the pressure relief mechanism **213** is provided on the bottom wall of the battery cell **20**, such that when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** can be discharged to the bottom of the battery **10**. In this way, the risk resulting from the emissions can be reduced by using the thermal management component at the bottom of the battery **10**, and the harm to users can be reduced because the bottom of the battery **10** is usually far away from the users.

(58) The pressure relief mechanism **213** may have various possible pressure relief structures, which is not limited by the embodiments of the present application. For example, the pressure relief mechanism **213** may be a temperature-sensitive pressure relief mechanism configured to be capable of being melted when the internal temperature of the battery cell **20** provided with the pressure relief mechanism **213** reaches a threshold; and/or the pressure relief mechanism **213** may be a pressure-sensitive pressure relief mechanism configured to be capable of being fractured when the internal gas pressure of the battery cell **20** provided with the pressure relief mechanism **213** reaches a threshold.

(59) FIG. **6** is a schematic diagram of a battery **10** according to an embodiment of the present application. As shown in FIG. **6**, the battery **10** may include the battery cell **20** and the thermal management component **13**.

(60) The battery cell **20** includes the pressure relief mechanism **213**, the pressure relief mechanism **213** is arranged in the first wall **21a** of the battery cell **20**, and the pressure relief mechanism **213** is configured, when the internal pressure or temperature of the battery cell **20** reaches a threshold, to be actuated to release the internal pressure or temperature. For example, the battery cell **20** may be the battery cell **20** in FIG. **5**.

(61) The thermal management component **13** is configured to accommodate a fluid to adjust the temperature of the plurality of battery cells **20**. In the case of lowering the temperature of the battery cells **20**, the thermal management component **13** may accommodate a cooling medium to adjust the temperature of the plurality of battery cells **20**. In this case, the thermal management component **13** may also be called a cooling component, a cooling system or a cooling plate, etc. In addition, the thermal management component **13** can also be used for heating, which is not limited by the embodiments of the present application. Optionally, the fluid can flow in a circulating manner to achieve better temperature adjustment effects.

(62) A first surface of the thermal management component **13** (an upper surface shown in FIG. **6**) is attached to the first wall **21a**. That is, the wall of the battery cell **20** provided with the pressure relief mechanism **213** is attached to the thermal management component **13**. The thermal management component **13** is configured to be capable of being damaged when the pressure relief mechanism **213** is actuated, so that the fluid is discharged from the inside of the thermal management component **13**.

(63) In an embodiment of the present application, the first surface of the thermal management component **13** is attached to the first wall **21a** provided with the pressure relief mechanism **213** therein. As such, when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** are discharged toward the thermal management component **13**. Also, the thermal management component **13** is configured to be capable of being damaged when the pressure relief mechanism **213** is actuated, so that the fluid is discharged from the inside of the thermal management component **13**. As such, the fluid can absorb the heat from the battery cell **20**, and reduces the temperature of the emissions. Due to the cooling of the fluid, the temperature of the emissions from the battery cell **20** may be reduced rapidly, so that the risk caused by the

abnormality of a single battery cell **20** can be suppressed in the first time, the possibility of battery explosion can be reduced, and the safety of the battery can be enhanced.

(64) Optionally, in an embodiment of the application, the thermal management component **13** may further be configured to be capable of being damaged by the emissions discharged from the battery cell **20** when the pressure relief mechanism **213** is actuated, so that the emissions pass through the thermal management component **13**.

(65) Specifically, when the pressure relief mechanism **213** is actuated, on the one hand, the emissions discharged from the battery cell **20** may damage the thermal management component **13** and pass through the thermal management component **13** to move away from the battery cell **20**, and on the other hand, the fluid is discharged from the inside of the thermal management component **13** while cooling the hot emissions. Since the temperature of the emissions is very high, no matter whether the fluid is used to heat or cool the battery cell **20**, the temperature of the fluid is lower than the temperature of the emissions, so that the emissions can be cooled. As such, the emissions may be cooled on the one hand, and are drained away through the thermal management component **13** on the other hand, and the risk caused thereby is reduced as much as possible, allowing for enhancing the safety of the battery.

(66) In the embodiments of the present application, various possible ways can be used to allow the thermal management component **13** to be damaged when the pressure relief mechanism **213** is actuated, as described below by way of examples.

(67) Optionally, as shown in FIG. 7, in an embodiment of the present application, the thermal management component **13** is provided with a recess **134** therein, and a side surface of the recess **134** is configured to be capable of being damaged by the emissions discharged from the battery cell **20**, so that the fluid is discharged from the inside of the thermal management component **13**. For example, the side surface of the recess **134** is configured to be capable of being broken through and/or melted by the emissions, so that the fluid is discharged from the inside of the thermal management component **13**.

(68) Optionally, in an embodiment of the present application, a bottom wall of the recess **134** is configured to be damaged by the emissions when the pressure relief mechanism **213** is actuated, so that the emissions pass through the thermal management component **13**. For example, the bottom wall of the recess **134** is configured to be capable of being broken through and/or melted by the emissions, so that the emissions pass through the thermal management component **13**.

(69) In the case of use of the recess **134**, when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** rush into the recess **134**. Since the bottom wall of the recess **134** is relatively weak, the emissions will damage the bottom wall of the recess **134** and pass through the thermal management component **13**. In addition, the emissions rushing into the recess **134** also melt the side face of the recess **134**, so that the fluid is discharged from the inside of the thermal management component **13**, thereby cooling the hot emissions.

(70) Optionally, in an embodiment of the present application, the radial dimension of the recess **134** gradually decreases in a direction away from the pressure relief mechanism **213**. That is, the side face of the recess **134** is an inclined plane. This can increase the contact area with the emissions and facilitate the damage by the emissions. For example, an inclination angle of the side face of the recess **134** (an included angle between the side face and the plane where the bottom wall is located) may be in the range from 15° to 85°.

(71) Optionally, as shown in FIG. 7, in an embodiment of the present application, in order to facilitate passage of the emissions through the thermal management component **13**, the bottom wall of the recess **134** is provided with a weakened zone **135**, and the weakened zone **135** is configured to be damaged by the emissions when the pressure relief mechanism **213** is actuated, so that the emissions pass through the weakened zone **135**.

(72) Optionally, the weakened zone **135** may be arranged opposite to the pressure relief mechanism **213**. As such, when the pressure relief mechanism **213** is actuated, the emissions can directly

impact on the weakened zone to open the weakened zone **135**.

(73) The weakened zone **135** can adopt various arrangements that facilitate damage by the emissions, which is not limited by the embodiments of the present application.

(74) Optionally, the recess **134** may be provided in the first surface, that is, the recess **134** is provided in a surface of the thermal management component **13** facing the first wall **21a**. That is, an opening surface of the recess **134** faces the first wall **21a**.

(75) It should be understood that the opening of the recess **134** may also face away from the first wall **21a**. In this case, the bottom wall of the recess **134** is also easily damaged by the emissions.

(76) The thermal management component **13** can form a fluid flow channel from a thermally conductive material. The fluid flows in the flow channel, and conducts heat through the thermally conductive material to adjust the temperature of the battery cell **20**. Optionally, the weakened zone may only have the thermally conductive material and no fluid to form a relatively thin thermally conductive material layer, which is easily damaged by the emissions. For example, the bottom wall of the recess **134** may be a thin thermally conductive material layer to form the weakened zone **135**.

(77) Optionally, as shown in FIGS. **8a** to **8c**, in an embodiment of the present application, the thermal management component **13** may include a first thermally conductive plate **131** and a second thermally conductive plate **132**. The first thermally conductive plate **131** and the second thermally conductive plate **132** form a flow channel **133** for accommodating a fluid. The first thermally conductive plate **131** is located between the first wall **21a** and the second thermally conductive plate **132** and is attached to the first wall **21a**. A first region **131a** of the first thermally conductive plate **131** is recessed toward the second thermally conductive plate **132** to form a recess **134**, and the first region **131a** is connected to the second thermally conductive plate **132**. In this way, a flow channel **133** is formed around the recess **134**, while no flow channel is formed in the bottom wall of the recess **134**, thus forming a weakened zone.

(78) Optionally, the first thermally conductive plate **131** or the second thermally conductive plate **132** at the bottom wall of the recess **134** may also be removed to form a thinner weakened zone. For example, as shown in FIG. **8c**, in an embodiment of the present application, the first region **131a** is provided with a first through hole **136**, and a radial dimension of the first through hole **136** is smaller than that of the recess **134**. That is, the first thermally conductive plate **131** at the bottom wall of the recess **134** is removed, and the connection between the first thermally conductive plate **131** and the second thermally conductive plate **132** is remained at the bottom edge of the recess **134** to form a flow channel **133** around the recess **134**.

(79) Optionally, the second thermally conductive plate **132** corresponding to the first through hole **136** may also be thinned. That is, the thickness of the second thermally conductive plate **132** corresponding to the first through hole **136** is smaller than that of the second thermally conductive plate **132** in other regions, such that the weakened zone is more easily damaged by the emissions. Optionally, a weakened recess may also be provided on the second thermally conductive plate **132** corresponding to the first through hole **136**.

(80) FIGS. **9a** to **9c** show schematic diagrams of the thermal management component **13**. As shown in FIGS. **9a** to **9c**, the first thermally conductive plate **131** is recessed to form a recess **134**, and a region of the second thermally conductive plate **132** corresponding to the recess **134** has no flow channel and is provided with a weakened recess **132a**. In this way, after the first thermally conductive plate **131** and the second thermally conductive plate **132** are connected to each other, a weakened zone is formed at the bottom wall of the recess **134**.

(81) It should be understood that the bottom wall of the recess **134** can be thinned by other thinning methods. For example, a blind hole or a stepped hole may be formed in the first region **131a** of the first thermally conductive plate **131**; and/or a blind hole is formed in the second thermally conductive plate **132**.

(82) Optionally, in an embodiment of the present application, the weakened zone **135** has a

thickness less than or equal to 3 mm. For example, the weakened zone **135** may have a thickness of 1 mm or less.

(83) In addition to the weakened zone **135** with a smaller thickness, a weakened zone **135** made of a low-melting-point material may also be used to facilitate the melting thereof by the emissions. That is, the weakened zone **135** can have a lower melting point than the rest of the thermal management component **13**. For example, the material of the weakened zone **135** has a melting point below 400° C.

(84) It should be understood that the weakened zone **135** may be configured to be made of a low-melting-point material and have a smaller thickness. That is, the foregoing two implementations may be implemented alone or in combination.

(85) When actuated, the pressure relief mechanism **213** is deformed to communicate the inside and outside of the battery cell **20**. For example, with respect to the pressure relief mechanism **213** using an indentation, when actuated, the pressure relief mechanism **213** is fractured at the indentation and opened toward two sides. Accordingly, the pressure relief mechanism **213** needs a certain deformation space. In an embodiment of the present application, the recess **134** is configured as an avoidance chamber for enabling the pressure relief mechanism **213** to be opened when the pressure relief mechanism **213** is actuated. The avoidance chamber provides a deformation space for the pressure relief mechanism **213**, such that the pressure relief mechanism **213** is deformed toward the thermal management component **13** and fractured.

(86) In the case of being used as an avoidance chamber, the recess **134** should be arranged such that it meets the condition that the pressure relief mechanism **213** can be opened when actuated. Specifically, a depth of the recess **134** is related to a size of the pressure relief mechanism **213**. As an embodiment of the present application, the recess **134** has a depth greater than 1 mm. For example, the recess **134** may have a depth of 3 mm or more than 3 mm, so as to further facilitate the opening of the pressure relief mechanism **213**. An area of the opening of the recess **134** is also related to an area of the pressure relief mechanism **213**. In order that the pressure relief mechanism **213** can be opened, a ratio of the area of the opening of the recess **134** to the area of the pressure relief mechanism **213** needs to be greater than a certain value. In addition, in order to facilitate the damage to the side face of the recess **134** by the emissions, the ratio of the area of the opening of the recess **134** to the area of the pressure relief mechanism **213** also needs to be less than a certain value. For example, the ratio of the area of the opening of the recess **134** to the area of the pressure relief mechanism **213** may range from 0.5 to 2.

(87) When the pressure relief mechanism **213** is provided at the first wall **21a** of the battery cell **20**, at least a portion of the pressure relief mechanism **213** may protrude outward from the first wall **21a**. This can facilitate the installation of the pressure relief mechanism **213** and ensure the internal space of the battery cell **20**. Optionally, as shown in FIG. **10**, in an embodiment of the present application, in the case where at least a portion of the pressure relief mechanism **213** protrudes outward from the first wall **21a**, the avoidance chamber may be configured to accommodate the at least portion of the pressure relief mechanism **213**. In this way, the first wall **21a** of the battery cell **20** can be closely attached to the surface of the thermal management component **13**, which facilitates the fixation of the battery cell **20** and can also save space and improve the thermal management efficiency. Moreover, when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** can be discharged toward the avoidance chamber and away from the battery cell **20**, thereby reducing the risk resulting from the emissions, so that the safety of the battery can be enhanced.

(88) Optionally, in an embodiment of the present application, a portion of the first wall **21a** around the pressure relief mechanism **213** protrudes outward, and the avoidance chamber is configured to accommodate the outward protruding portion of the first wall **21a** around the pressure relief mechanism **213**. Similarly, in the case where the portion of the first wall **21a** around the pressure relief mechanism **213** protrudes outward, the avoidance chamber can ensure that the first wall **21a**

of the battery cell **20** can be closely attached to the surface of the thermal management component **13**, which facilitates the fixation of the battery cell **20** and can also save space and improve the thermal management efficiency.

(89) In the above embodiments, provision of the recess **134** makes it possible for the thermal management component **13** to be damaged when the pressure relief mechanism **213** is actuated. In addition to the recess **134**, a through hole may also be provided to make it possible for the thermal management component **13** to be damaged when the pressure relief mechanism **213** is actuated.

(90) Optionally, as shown in FIG. **11**, in an embodiment of the present application, the thermal management component **13** is provided with a second through hole **137**, and the second through hole **137** is configured such that the emissions discharged from the battery cell **20** can pass through the thermal management component **13** via the second through hole **137** when the pressure relief mechanism **213** is actuated.

(91) Optionally, the second through hole **137** may be arranged opposite to the pressure relief mechanism **213**. As such, when the pressure relief mechanism **213** is actuated, the emissions may directly pass through thermal management component **13** via the second through hole **137**.

(92) Optionally, in an embodiment of the present application, in the case of provision of the second through hole **137**, the part of the thermal management component **13** around the second through hole **137** can be damaged by the emissions from the battery cell **20**, so that the fluid is discharged from the inside of the thermal management component **13**.

(93) Specifically, when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** pass through thermal management component **13** via the second through hole **137**. In addition, the emissions also damage portions around the second through hole **137**. For example, the hot emissions melt the surrounding thermal management component **13**, so that the fluid is discharged from the inside of the thermal management component **13**, thereby cooling the hot emissions.

(94) Optionally, in an embodiment of the present application, the hole wall of the second through hole **137** is configured to be capable of being damaged by the emissions, so that the fluid is discharged from the inside of the thermal management component **13**.

(95) When the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** rush into the second through hole **137**. Since the emissions have high pressure and high temperature, the emissions further melt the hole wall of the second through hole **137** when passing through the second through hole **137**, so that the fluid is discharged from the inside of the thermal management component **13**, thereby cooling the emissions.

(96) Optionally, the radial dimension of the second through hole **137** gradually decreases in the direction away from the pressure relief mechanism **213**. That is, the hole wall of the second through hole **137** is an inclined plane. This can increase the contact area with the emissions and facilitate the damage by the emissions.

(97) Optionally, similar to the foregoing recess **134**, the second through hole **137** should also be arranged such that it meets the condition that the pressure relief mechanism **213** can be opened when actuated. As an embodiment of the present application, an area of an opening of the second through hole **137** is related to an area of the pressure relief mechanism **213**. In order that the pressure relief mechanism **213** can be opened, a ratio of the area of the opening of the second through hole **137** to the area of the pressure relief mechanism **213** needs to be greater than a certain value. In addition, in order to facilitate damage to the hole wall of the second through hole **137** by the emissions, the ratio of the opening area of the second through hole **137** to the area of the pressure relief mechanism **213** should also be less than a certain value. For example, the ratio of the area of the opening of the second through hole **137** to the area of the pressure relief mechanism **213** ranges from 0.5 to 2.

(98) Optionally, in an embodiment of the present application, the pressure relief mechanism **213** is arranged in the first wall **21a** of the battery cell **20** provided with the pressure relief mechanism

213, the first wall **21a** is attached to thermal management component **13**, at least a portion of the pressure relief mechanism **213** protrudes outward from the first wall **21a**, and the second through hole **137** is used to accommodate the at least portion of the pressure relief mechanism **213**. In this way, the first wall **21a** of the battery cell **20** can be closely attached to the surface of the thermal management component **13**, which facilitates the fixation of the battery cell **20** and can also save space and improve the thermal management efficiency. Moreover, when the pressure relief mechanism **213** is actuated, the emissions from the battery cell **20** can be discharged toward the second through hole **137** and away from the battery cell **20**, thereby reducing the risk resulting from the emissions, so that the safety of the battery can be enhanced.

(99) Optionally, in an embodiment of the present application, a portion of the first wall **21a** around the pressure relief mechanism **213** protrudes outward, and the second through hole **137** is configured to accommodate an outward protruding portion of the first wall **21a** around the pressure relief mechanism **213**. In the case that the portion of the first wall **21a** around the pressure relief mechanism **213** protrudes outward, the second through hole **137** may ensure that the first wall **21a** of the battery cell **20** may be closely attached to the surface of the thermal management component **13**, which is easy to fix the battery cell **20**, saves space and improves the thermal management efficiency.

(100) It should be understood that, in addition to providing the thermal management component **13** with a structure such that the thermal management component **13** can be damaged when the pressure relief mechanism **213** is actuated, the pressure relief mechanism **213** may be also provided with a structure that enables the thermal management component **13** to be damaged when the pressure relief mechanism **213** is actuated.

(101) Optionally, in an embodiment of the present application, the pressure relief mechanism **213** is provided with a breaking device. The breaking device is configured to damage the thermal management component **13** when the pressure relief mechanism **213** is actuated, such that the fluid is discharged from the inside of the thermal management component **13**. For example, the breaking device may be a spike, but this is not limited by the embodiment of the present application.

(102) Optionally, in an embodiment of the present application, as shown in FIG. **12**, the battery **10** may further include an electrical chamber **11a** and a collection chamber **11b**. The thermal management component **13** is configured to isolate the electrical chamber **11a** from the collection chamber **11b**. The so-called “isolate” here refers to separate, which may or may not be sealed.

(103) The electrical chamber **11a** is used to accommodate a plurality of the battery cells **20**. The electrical chamber **11a** may also be used to accommodate the bus component **12**. The electrical chamber **11a** provides an accommodation space for the battery cells **20** and the bus component **12**, and the electrical chamber **11a** may be shaped according to the plurality of battery cells **20** and the bus component **12**.

(104) The bus component **12** is configured to electrically connect the plurality of battery cells **20**. The bus component **12** may implement the electrical connection between the battery cells **20** by connecting electrode terminals **214** of the battery cells **20**.

(105) The collection chamber **11b** is used to collect the emissions discharged from the battery cell **20** and the emissions from the thermal management component **13** when the pressure relief mechanism **213** is actuated.

(106) In the embodiment of the present application, the thermal management component **13** is used to isolate the electrical chamber **11a** from the collection chamber **11b**. That is, the electrical chamber **11a** for accommodating the plurality of battery cells **20** and the bus component **12** is separated from the collection chamber **11b** for collecting the emissions. In this way, when the pressure relief mechanism **213** is actuated, the emissions from the battery cells **20** enter the collection chamber **11b** rather than the electrical chamber, or a small amount of emissions enter the electrical chamber **11a**, so that the electrical connection in the electrical chamber **11a** is not affected, and therefore the safety of the battery can be enhanced.

(107) Optionally, in an embodiment of the application, the thermal management component **13** is configured to make it possible for the emissions discharged from the battery cell **20** to pass through the thermal management component **13** and enter the collection chamber **11b** when the pressure relief mechanism **213** is actuated.

(108) Optionally, in an embodiment of the present application, the thermal management component **13** has a wall shared by the electrical chamber **11a** and the collection chamber **11b**. As shown in FIG. **12**, the thermal management component **13** may be both a wall of the electrical chamber **11a** and a wall of the collection chamber **11b**. That is, the thermal management component **13** (or a portion thereof) can be directly used as a wall shared by the electrical chamber **11a** and the collection chamber **11b**. In this way, the emissions from the battery cell **20** can enter the collection chamber **11b** through the thermal management component **13**. Besides, due to the existence of the thermal management component **13**, the emissions can be isolated from the electrical chamber **11a** as far as possible, thus reducing the risk resulting from the emissions and enhancing the safety of the battery.

(109) Optionally, in an embodiment of the present application, the electrical chamber **11a** may be composed of a covering having an opening and a thermal management component **13**. For example, as shown in FIG. **13**, a covering **110** has an opening (a lower side opening in FIG. **13**). The covering **110** with the opening is a semi-closed chamber with an opening in communication with the outside, and the thermal management component **13** covers the opening to form a chamber, i.e., an electrical chamber **11a**.

(110) Optionally, the covering **110** may be composed of multiple portions. For example, as shown in FIG. **14**, the covering **110** may include a first portion **111** and a second portion **112**. Two sides of the second portion **112** have openings, respectively. The first portion **111** covers the opening on one side of the second portion **112**, and the thermal management component **13** covers the opening on the other side of the second portion **112**, thus forming the electrical chamber **11a**.

(111) The embodiment of FIG. **14** may be obtained through improvements on the basis of FIG. **2**. Specifically, a bottom wall of the second portion **112** in FIG. **2** may be replaced with the thermal management component **13**, and the thermal management component **13** acts as a wall of the electrical chamber **11a**, thus forming the electrical chamber **11a** in FIG. **14**. In other words, the bottom wall of the second portion **112** in FIG. **2** can be removed. That is, an annular wall with two opening sides is formed, and the first portion **111** and the thermal management component **13** cover the openings on the two sides of the second portion **112** respectively to form a chamber, namely an electrical chamber **11a**.

(112) Optionally, in an embodiment of the present application, the collection chamber **11b** may be composed of a thermal management component **13** and a protective member. For example, as shown in FIG. **15**, the battery **10** further includes a protective member **115**. The protective member **115** is configured to protect the thermal management component **13**, and the protective member **115** and the thermal management component **13** form the collection chamber **11b**.

(113) The collection chamber **11b** formed by the protective member **115** and the thermal management component **13** does not occupy the space that may accommodate the battery cell. Therefore, the collection chamber **11b** with a larger space therein can be provided, which may effectively collect and buffer the emissions and reduce the risk resulting thereof.

(114) Optionally, in an embodiment of the present application, the fluid, such as a cooling medium, or a component for accommodating the fluid, may be further provided in the collection chamber **11b** to further cool the emissions entering the collection chamber **11b**.

(115) Optionally, in an embodiment of the present application, the collection chamber **11b** may be a sealed chamber. For example, the connection between the protective member **115** and the thermal management component **13** may be sealed by a sealing member.

(116) Optionally, in an embodiment of the present application, the collection chamber **11b** may not be a sealed chamber. For example, the collection chamber **11b** may be in communication with the

air, and as such, a part of the emissions may further be discharged to the outside of the collection chamber **11b**.

(117) In the foregoing embodiment, the thermal management component **13** covers the opening of the covering **110** to form an electrical chamber **11a**, and the thermal management component **13** and the protective member **115** form the collection chamber **11b**. Optionally, the thermal management component **13** may also directly separate the closed covering into the electrical chamber **11a** and the collection chamber **11b**.

(118) For example, as shown in FIG. **16**, in an embodiment of the present application, the thermal management component **13** is arranged inside the covering **110**, and separate the interior of the covering **110** into the electrical chamber **11a** and the collection chamber **11b**. That is, the closed covering **110** internally forms a chamber, and the thermal management component **13** separates the chamber inside the covering **110** into two chambers, namely the electrical chamber **11a** and the collection chamber **11b**.

(119) Since the electrical chamber **11a** needs a relatively large space to accommodate a plurality of battery cells **20**, etc., the thermal management component **13** may be provided near a certain wall of the covering **110** to isolate the electrical chamber **11a** with a relatively large space from the collection chamber **11b** with a relatively small space.

(120) Optionally, as shown in FIG. **17**, in an embodiment of the present application, the covering **110** may include a first portion **111** and a second portion **112**. A side of the second portion **112** has an opening to form a semi-closed structure. The semi-closed structure is a chamber with an opening. The thermal management component **13** is provided inside the second portion **112**, and the first portion **111** covers the opening of the second portion **112**. In other words, the thermal management component **13** can be first placed in the semi-closed second portion **112** to isolate the collection chamber **11b**, and then the first portion **111** covers the opening of the second portion **112** to form the electrical chamber **11a**.

(121) Optionally, in an embodiment of the present application, the electrical chamber **11a** is isolated from the collection chamber **11b** by the thermal management component **13**. That is, the collection chamber **11b** is not in communication with the electrical chamber **11a**, and liquid or gas, etc. in the collection chamber **11b** cannot enter the electrical chamber **11a**, so that the electrical chamber **11a** can be better protected.

(122) FIG. **18** is an exploded view of a battery **10** according to an embodiment of the present application. In the embodiment shown in FIG. **18**, the thermal management component **13** is provided with a recess **134**, and forms a collection chamber together with a protective member **115**.

(123) For the description of each component in the battery **10**, reference can be made to the foregoing embodiments, which will not be repeated here for brevity.

(124) An embodiment of the present application further provides a power consumption device, which may include the battery **10** in each of the foregoing embodiments. Optionally, the power consumption device may be a vehicle **1**, a ship or a spacecraft.

(125) The battery and the power consumption device of the embodiments of the present application are described above, and a method and a device for preparing a battery of the embodiments of the present application will be described below. For the parts that are not described in detail, reference is made to the foregoing embodiments.

(126) FIG. **19** shows a schematic flowchart of a method **300** for preparing a battery according to an embodiment of the present application. As shown in FIG. **19**, the method **300** may include: **310**, providing a battery cell **20**, the battery cell **20** including a pressure relief mechanism **213**, the pressure relief mechanism **213** being arranged in a first wall **21a** of the battery cell **20**, and the pressure relief mechanism **213** being configured, when an internal pressure or temperature of the battery cell **20** reaches a threshold, to be actuated to release the internal pressure or temperature; **320**, providing a thermal management component **13**, the thermal management component **13** being used to contain a fluid; and **330**, attaching a first surface of the thermal management

component **13** to the first wall **21a**, wherein the thermal management component **13** can be damaged when the pressure relief mechanism **213** is actuated, so that the fluid is discharged from inside of the thermal management component **13**.

(127) FIG. **20** shows a schematic block diagram of a device **400** for preparing a battery according to an embodiment of the present application. As shown in FIG. **20**, the device **400** for preparing a battery may include: a provision module **410** and an installation module **420**.

(128) The provision module **410** is configured to: provide a battery cell **20**, the battery cell **20** including a pressure relief mechanism **213**, the pressure relief mechanism **213** being arranged in a first wall **21a** of the battery cell **20**, and the pressure relief mechanism **213** being configured, when an internal pressure or temperature of the battery cell **20** reaches a threshold, to be actuated to release the internal pressure or temperature; and provide a thermal management component **13**, the thermal management component **13** being used to contain a fluid.

(129) The installation module **420** is configured to: attach a first surface of the thermal management component **13** to the first wall **21a**, wherein the thermal management component **13** can be damaged when the pressure relief mechanism **213** is actuated, so that the fluid is discharged from inside of the thermal management component **13**.

(130) It should be finally noted that, the above embodiments are merely used for illustrating rather than limiting the technical solutions of the present application. Although the present application is illustrated in detail with reference to the foregoing embodiments, those of ordinary skill in the art should understand that they can still modify the technical solutions described in the foregoing embodiments, or make equivalent substitutions to some of the technical features therein, but these modifications or substitutions can be made to the respective technical solutions without departing from the spirit and scope of the technical solutions of the embodiments of the present application.

Claims

1. A battery, comprising: a battery cell comprising a pressure relief mechanism, the pressure relief mechanism being arranged in a first wall of the battery cell, and the pressure relief mechanism being configured, when an internal pressure or temperature of the battery cell reaches a threshold, to be actuated to release the internal pressure; and a thermal management component for containing a fluid to adjust a temperature of the battery cell; wherein a first surface of the thermal management component is attached to the first wall, when the pressure relief mechanism is actuated, emissions from the battery cell are discharged toward the thermal management component, and the thermal management component is configured to be capable of being damaged when the pressure relief mechanism is actuated, so that the fluid is discharged from inside of the thermal management component; wherein the thermal management component is provided with a second through hole therein, the second through hole being configured such that emissions discharged from the battery cell can pass through the thermal management component via the second through hole when the pressure relief mechanism is actuated; wherein a radial dimension of the second through hole gradually decreases in a direction away from the pressure relief mechanism.
2. The battery according to claim 1, wherein the thermal management component is configured to be damaged by emissions discharged from the battery cell when the pressure relief mechanism is actuated, so that the emissions pass through the thermal management component.
3. The battery according to claim 2, wherein the hole wall of the second through hole is configured to be capable of being broken through and/or melted by the emissions, so that the fluid is discharged from the inside of the thermal management component.
4. The battery according to claim 1, wherein a hole wall of the second through hole is configured to be capable of being damaged by the emissions, so that the fluid is discharged from the inside of the thermal management component.
5. The battery according to claim 1, wherein an area of an opening of the second through hole is

related to an area of the pressure relief mechanism.

6. The battery according to claim 5, wherein a ratio of the area of the opening of the second through hole to the area of the pressure relief mechanism ranges from 0.5 to 2.

7. The battery according to claim 1, wherein at least a portion of the pressure relief mechanism protrudes from the first wall, and the second through hole is used to accommodate the at least portion of the pressure relief mechanism.

8. The battery according to claim 1, wherein a portion of the first wall around the pressure relief mechanism protrudes outward, and the second through hole is used to accommodate an outward protruding portion of the first wall around the pressure relief mechanism.

9. The battery according to claim 1, wherein the pressure relief mechanism is provided with a breaking device, the breaking device being used to damage the thermal management component when the pressure relief mechanism is actuated, so that the fluid is discharged from the inside of the thermal management component.

10. The battery according to claim 9, wherein the breaking device is a spike.

11. The battery according to claim 1, wherein the second through hole is arranged opposite to the pressure relief mechanism.

12. The battery according to claim 1, wherein a part of the thermal management component around the second through hole can be damaged by the emissions from the battery cell, so that the fluid is discharged from the inside of the thermal management component.

13. The battery according to claim 1, further comprising: an electrical chamber for accommodating a plurality of the battery cells; and a collection chamber for collecting emissions discharged from the battery cells and emissions from the thermal management component when the pressure relief mechanism is actuated; wherein the thermal management component is configured to isolate the electrical chamber from the collection chamber, and the thermal management component has a wall shared by the electrical chamber and the collection chamber; and the thermal management component is configured to enable the emissions discharged from the battery cell to pass through the thermal management component and enter the collection chamber when the pressure relief mechanism is actuated.

14. A power consumption device, comprising: a battery; wherein the battery comprises: a battery cell comprising a pressure relief mechanism, the pressure relief mechanism being arranged in a first wall of the battery cell, and the pressure relief mechanism being configured, when an internal pressure or temperature of the battery cell reaches a threshold, to be actuated to release the internal pressure; and a thermal management component for containing a fluid to adjust a temperature of the battery cell; wherein a first surface of the thermal management component is attached to the first wall, when the pressure relief mechanism is actuated, emissions from the battery cell are discharged toward the thermal management component, and the thermal management component is configured to be capable of being damaged when the pressure relief mechanism is actuated, so that the fluid is discharged from inside of the thermal management component; wherein the thermal management component is provided with a second through hole therein, the second through hole being configured such that emissions discharged from the battery cell can pass through the thermal management component via the second through hole when the pressure relief mechanism is actuated; wherein a radial dimension of the second through hole gradually decreases in a direction away from the pressure relief mechanism.
